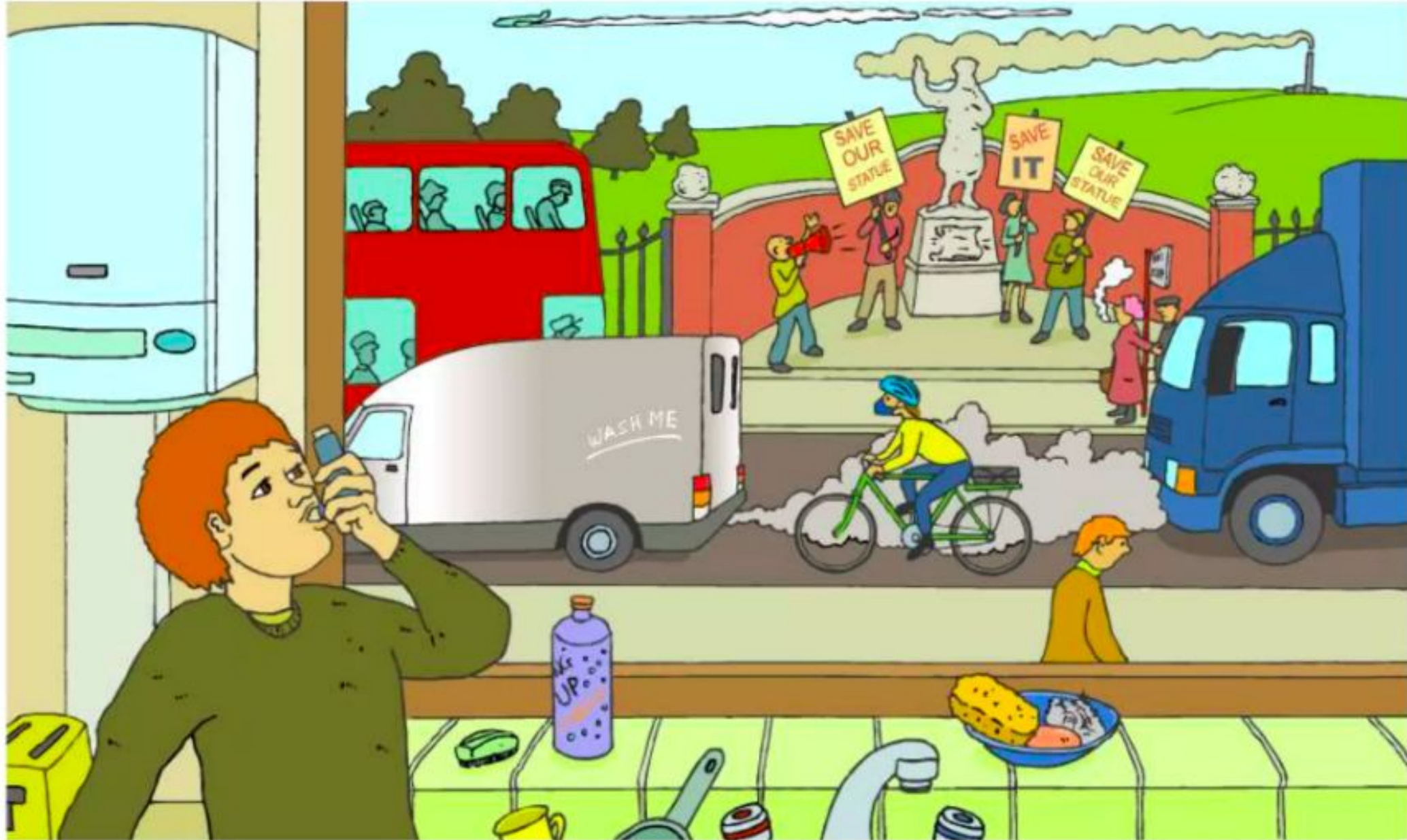


Starter

How do human activities affect the air we breathe?



Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
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Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

Title: Atmosphere and Climate Change

LO: Today we are learning about why global warming happens and its impacts.

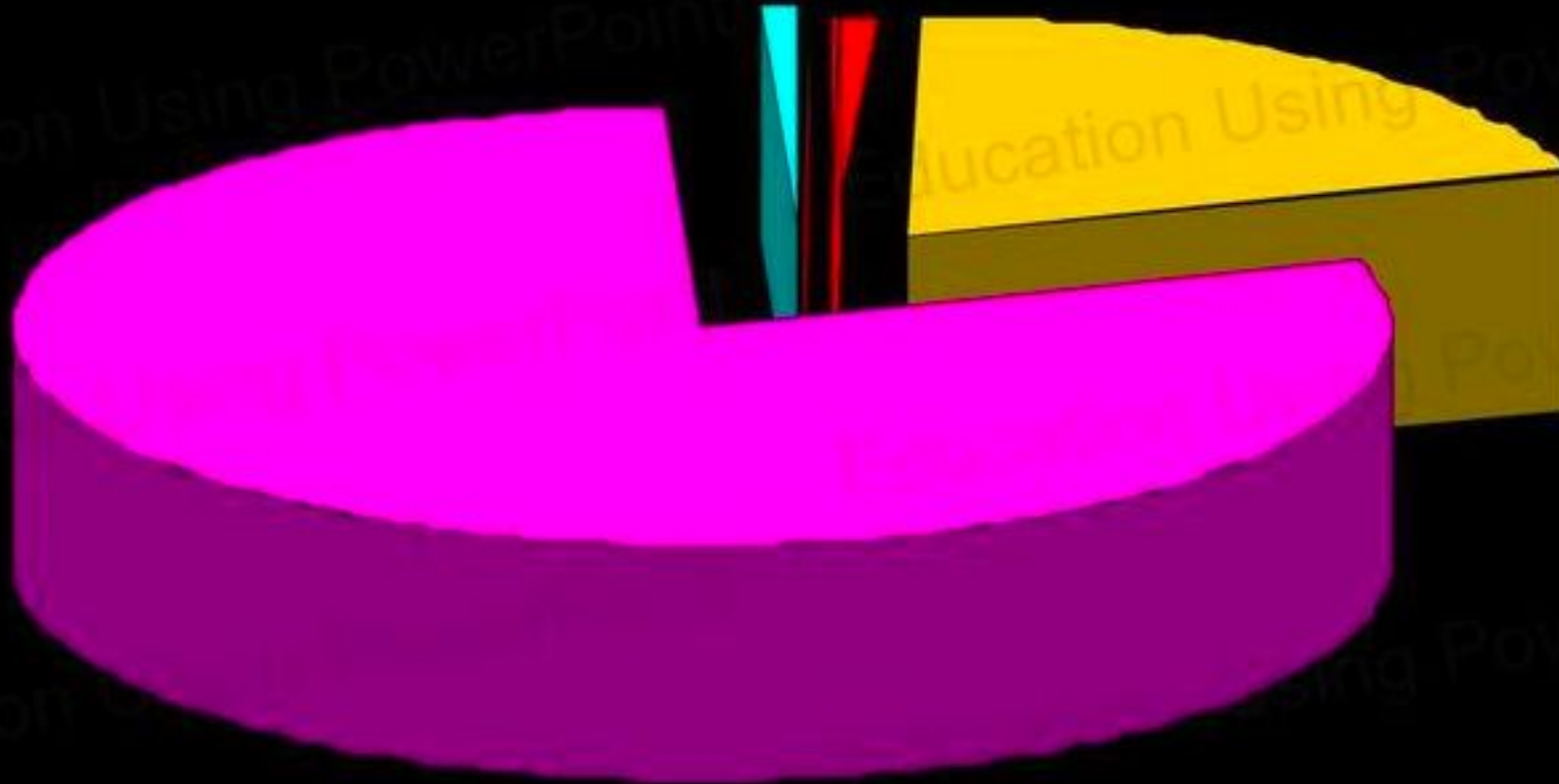
After this topic you will be able to:

- Explain why global warming happens.
- Identify the gases responsible and how they are produced.
- Explain some impacts of global warming.

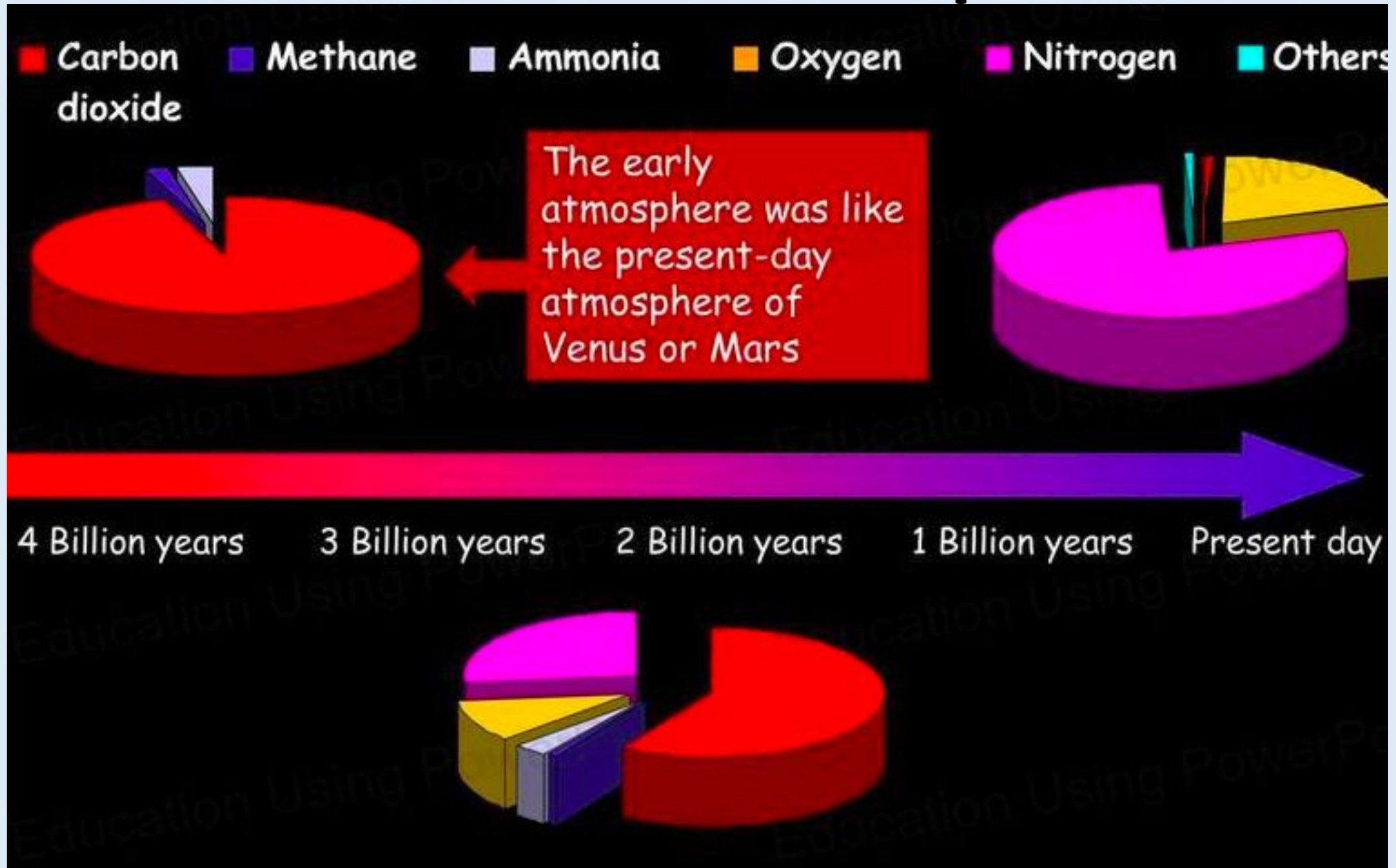
The Earth's Atmosphere

For the last 200 million years the atmosphere has remained roughly the same - it contains 78% nitrogen, 21% oxygen, 1% noble gases and about 0.03% CO_2

■ Carbon dioxide, water vapour ■ Oxygen ■ Nitrogen ■ Noble gases

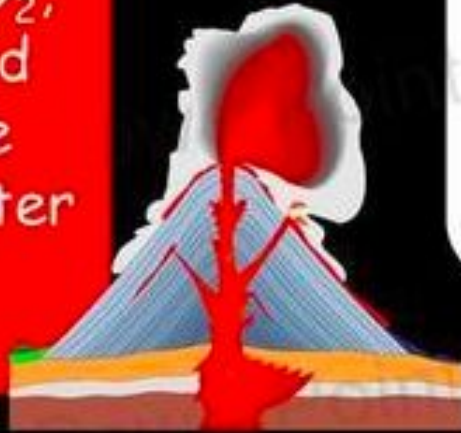


Evolution of the Earth's Atmosphere



Evolution of the Earth's Atmosphere

Volcanic activity releases nitrogen, CO_2 , methane, ammonia and water vapour into the atmosphere. The water vapour condenses to form oceans.



Evidence for the early atmosphere is limited. Why is that?



4 Billion years

3 Billion years

2 Billion years

1 Billion years

Present day

Carbon from CO_2 becomes locked up in sedimentary rocks as carbonates and fossil fuels. As the oceans form by condensing water vapour the carbonates and fossil fuels become locked underground, decreasing the amount of CO_2 .

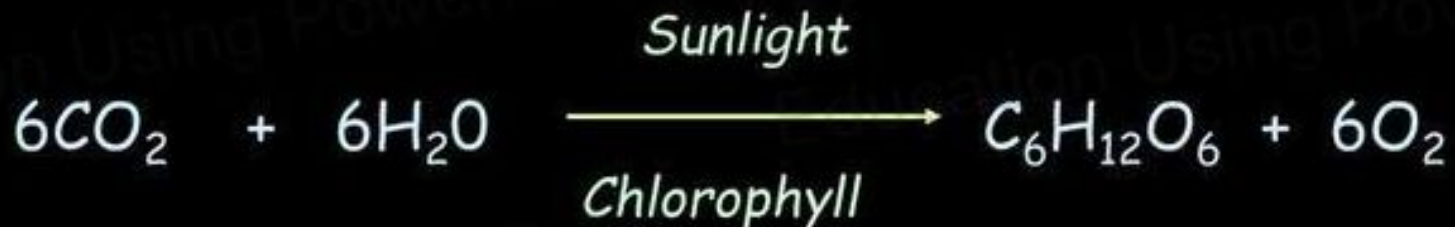
Changes in Oxygen and Carbon Dioxide

Somewhere around 2.7 billion years ago, algae and plants arrived and started taking in carbon dioxide and producing oxygen. A billion years later, there was enough oxygen for animals to evolve.



Q. What are the equations for photosynthesis?

Carbon dioxide + water $\xrightarrow[\text{Chlorophyll}]{\text{Sunlight}}$ glucose + oxygen



Decreasing Carbon Dioxide

As well as photosynthesis by plants and algae, atmospheric carbon dioxide levels were also decreased by the formation of carbonate rocks, e.g.:



Limestone

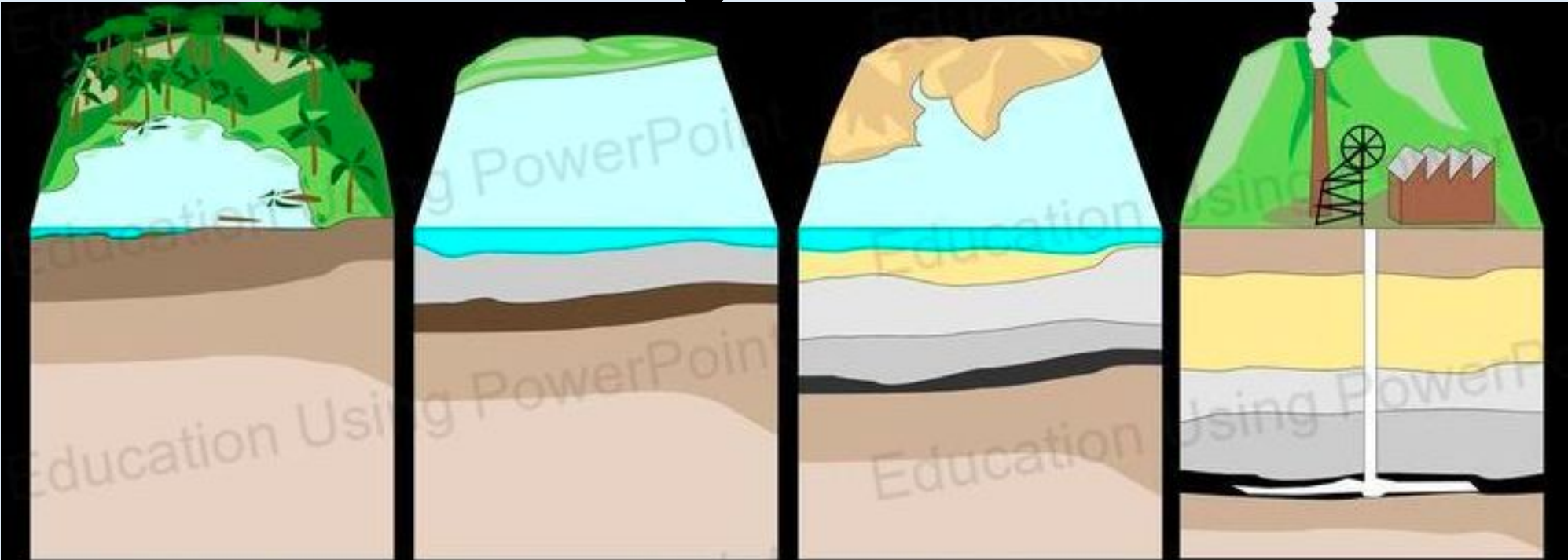


Chalk



Marble (made from
chalk or limestone)

Formation of oil and gas



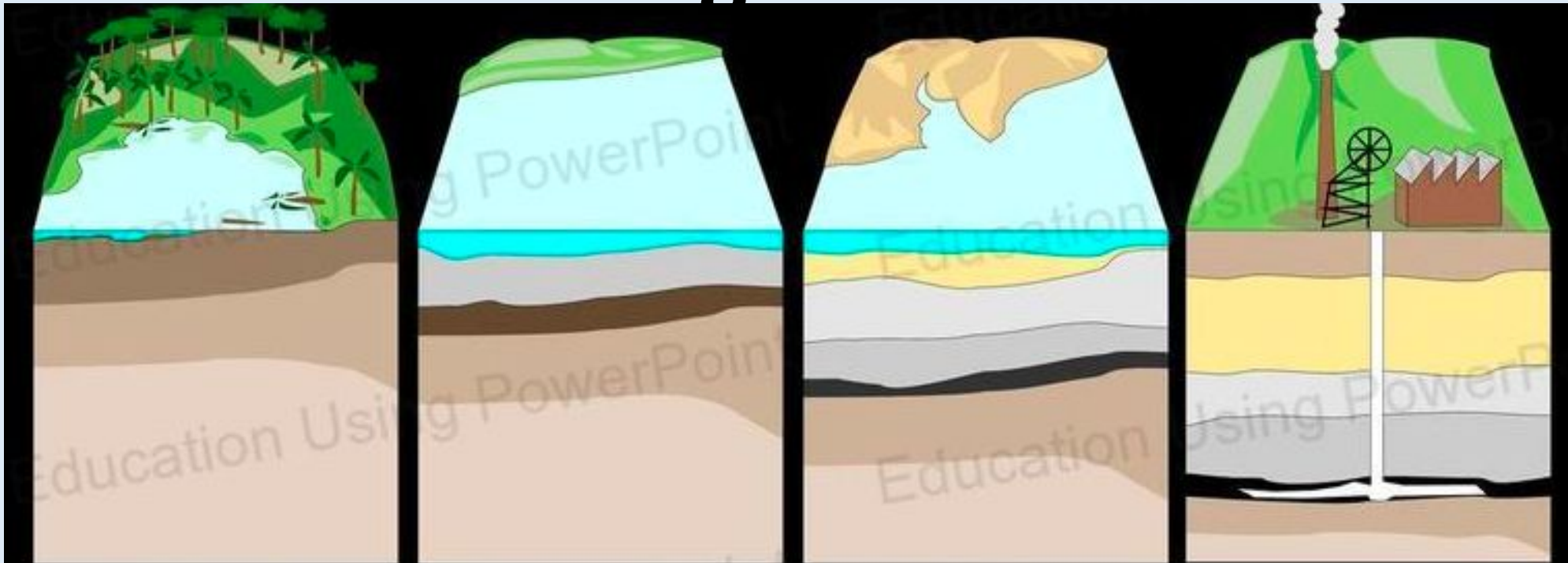
1) Layers of dead sea _____ settle on the seabed.

2) Layers of _____ rock build up on top.

3) The heat and _____ from these rocks, along with the absence of _____, mean that oil and gas are formed over _____ of years.

Words - sedimentary, millions, oxygen, creatures, pressure

Formation of oil and gas



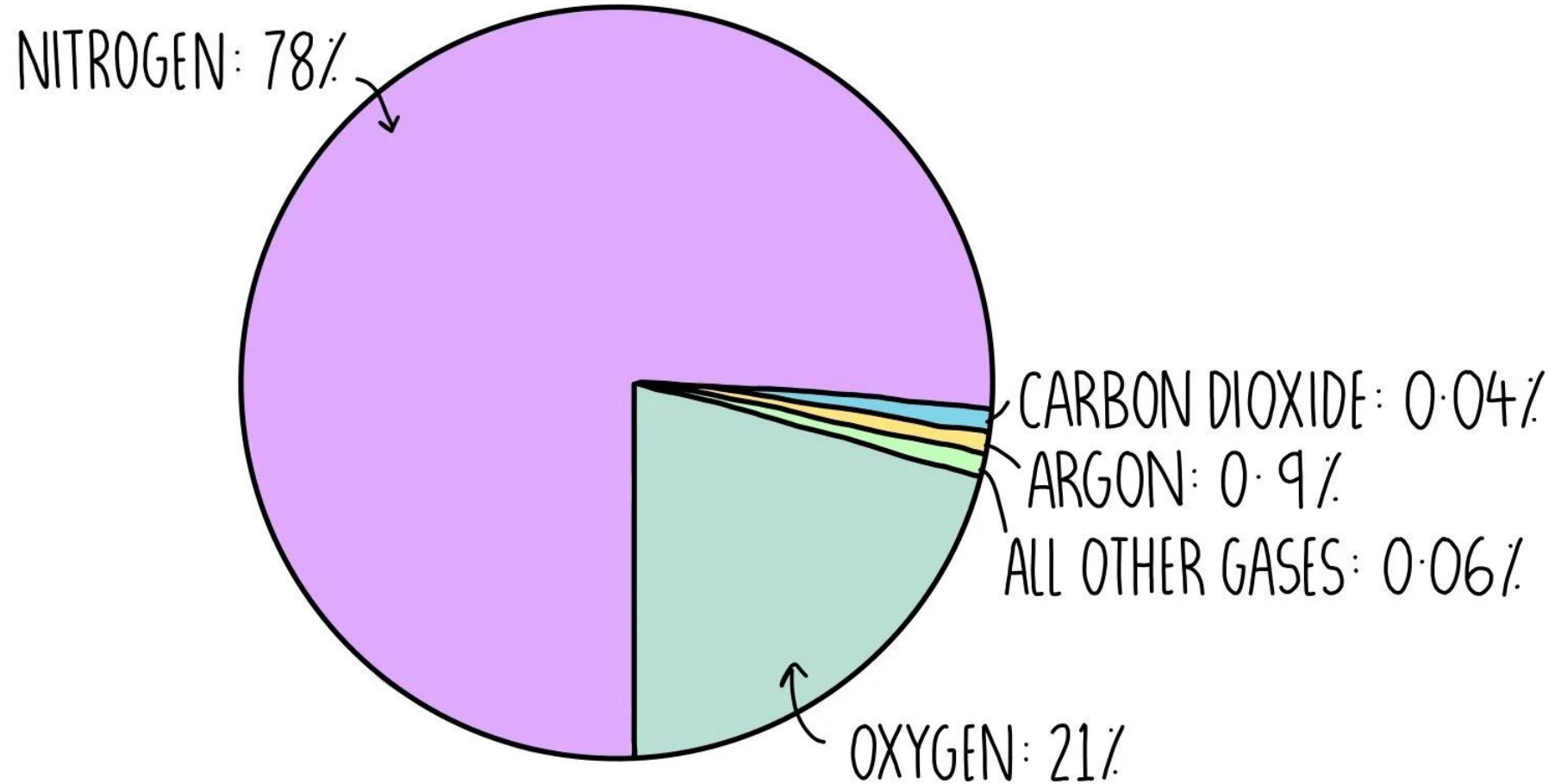
1) Layers of dead sea creatures settle on the seabed.

2) Layers of sedimentary rock build up on top.

3) The heat and pressure from these rocks, along with the absence of oxygen, mean that oil and gas are formed over millions of years.

Words - sedimentary, millions, oxygen, creatures, pressure

Percentage of gases in the atmosphere

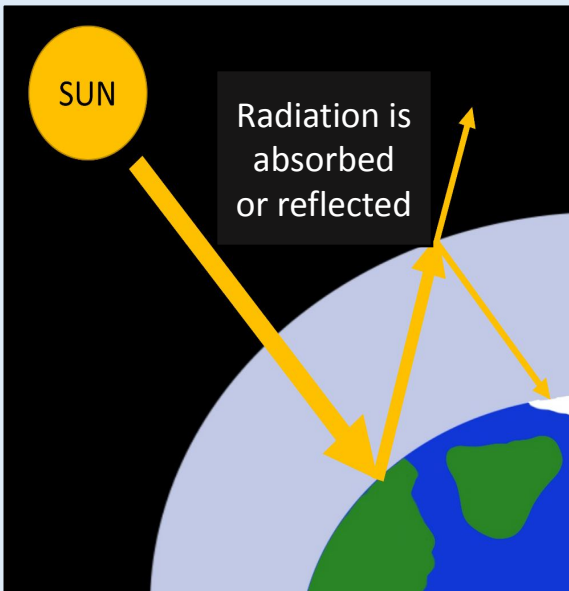


Group Reading

10:00

Reader 1:

The temperature of our planet is rising. This is **global warming**. The world temperature has already increased 1°C* since before we started burning fossil fuels on a large scale and it is already causing problems. We are experiencing more droughts, flooding and heatwaves. These changes may cause the extinction of some species and make it harder to grow enough food and store enough water.



Reader 2:

When **radiation** from the sun reaches Earth some is absorbed and some reflects back into space. Greenhouse gases, like methane and carbon dioxide, absorb some of the energy and radiate it back toward the Earth. This is called the **greenhouse effect** and it makes the Earth warmer than it would be without them. We need some greenhouse effect. Without it the Earth would be too cold to support us.

Group Reading

Reader 3:

We have been releasing more greenhouse gases into the atmosphere since the Industrial Revolution. The main sources of these **emissions** are carbon dioxide from the combustion of fossil fuels and methane from farming animals.



Reader 4:

Deforestation is making the problem worse. Deforestation is when humans cut down trees to make room for farmland, new roads or buildings. This means that there are less trees to remove CO_2 from the atmosphere through photosynthesis. If the trees burn or rot, this also adds the CO_2 that was stored in them into the atmosphere.



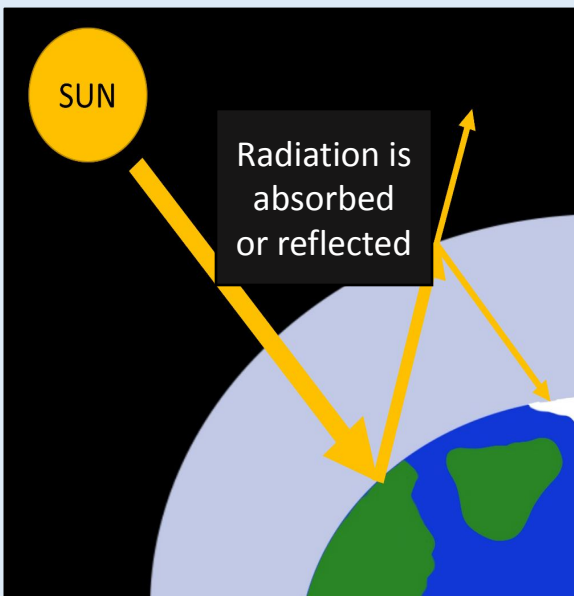
10:00

1. What is global warming and what has caused the temperature to increase?
2. What problems are happening because of global warming?
3. How do greenhouse gases affect the radiation from the sun?
4. Why do we need the greenhouse effect for life on Earth?
5. What human activities have increased greenhouse gases since the Industrial Revolution?
6. What are the main sources of carbon dioxide and methane?
7. What is deforestation and why do humans cut down trees?

Mandarin - Group Reading

Reader 1:

地球温度正在上升，这就是全球变暖。自我们开始大规模燃烧化石燃料以来，全球气温已经上升了1摄氏度*，并且已经造成了诸多问题。我们正经历着更频繁的干旱、洪水和热浪。这些变化可能导致一些物种灭绝，并使粮食生产和水资源储存变得更加困难。



Reader 2:

当太阳辐射到达地球时，一部分被吸收，一部分反射回太空。温室气体，例如甲烷和二氧化碳，会吸收部分能量并将其辐射回地球。这被称为温室效应，它使地球比没有温室气体时更温暖。我们需要一定的温室效应。如果没有温室效应，地球会太冷，无法维持我们的生存。

Mandarin - Group Reading

Reader 3:

自工业革命以来，我们一直在向大气中排放更多的温室气体。这些排放物的主要来源是化石燃料燃烧产生的二氧化碳和畜牧业产生的甲烷。



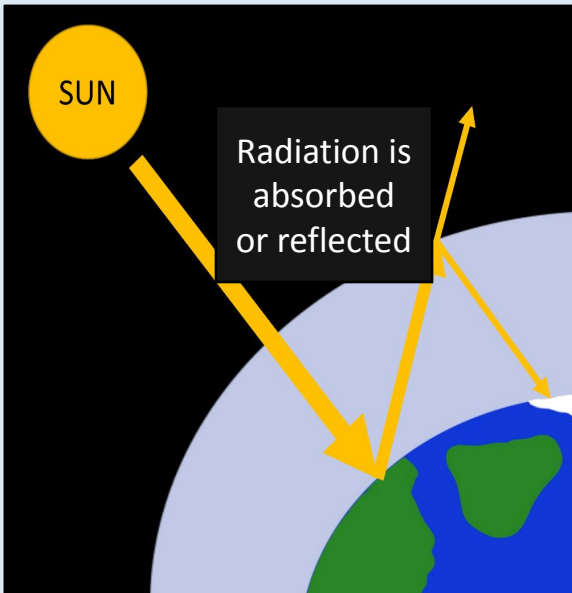
Reader 4:

森林砍伐加剧了这个问题。森林砍伐是指人类为了开垦农田、修建道路或建造房屋而砍伐树木。这意味着通过光合作用从大气中吸收二氧化碳的树木减少了。如果树木被烧毁或腐烂，也会将储存在树中的二氧化碳释放到大气中。

Korean - Group Reading

Reader 1:

지구의 온도가 상승하고 있습니다. 이것이 바로 지구 온난화입니다. 화석 연료를 대량으로 사용하기 시작하기 전보다 지구 온도는 이미 1°C^* 상승했으며, 이는 이미 문제를 일으키고 있습니다. 우리는 더 많은 가뭄, 홍수, 그리고 폭염을 겪고 있습니다. 이러한 변화는 일부 종의 멸종을 초래할 수 있으며, 충분한 식량을 재배하고 충분한 물을 저장하는 것을 더욱 어렵게 만들 수 있습니다.



Reader 2:

태양 복사 에너지가 지구에 도달하면 일부는 흡수되고 일부는 우주로 반사됩니다. 메탄이나 이산화탄소와 같은 온실가스는 에너지의 일부를 흡수하여 지구로 다시 방출합니다. 이를 온실 효과라고 하며, 온실가스가 없다면 지구는 더 따뜻해질 것입니다. 우리에게 온실 효과가 필요합니다. 온실 효과가 없다면 지구는 너무 차가워져 우리를 지탱할 수 없을 것입니다.

Korean - Group Reading

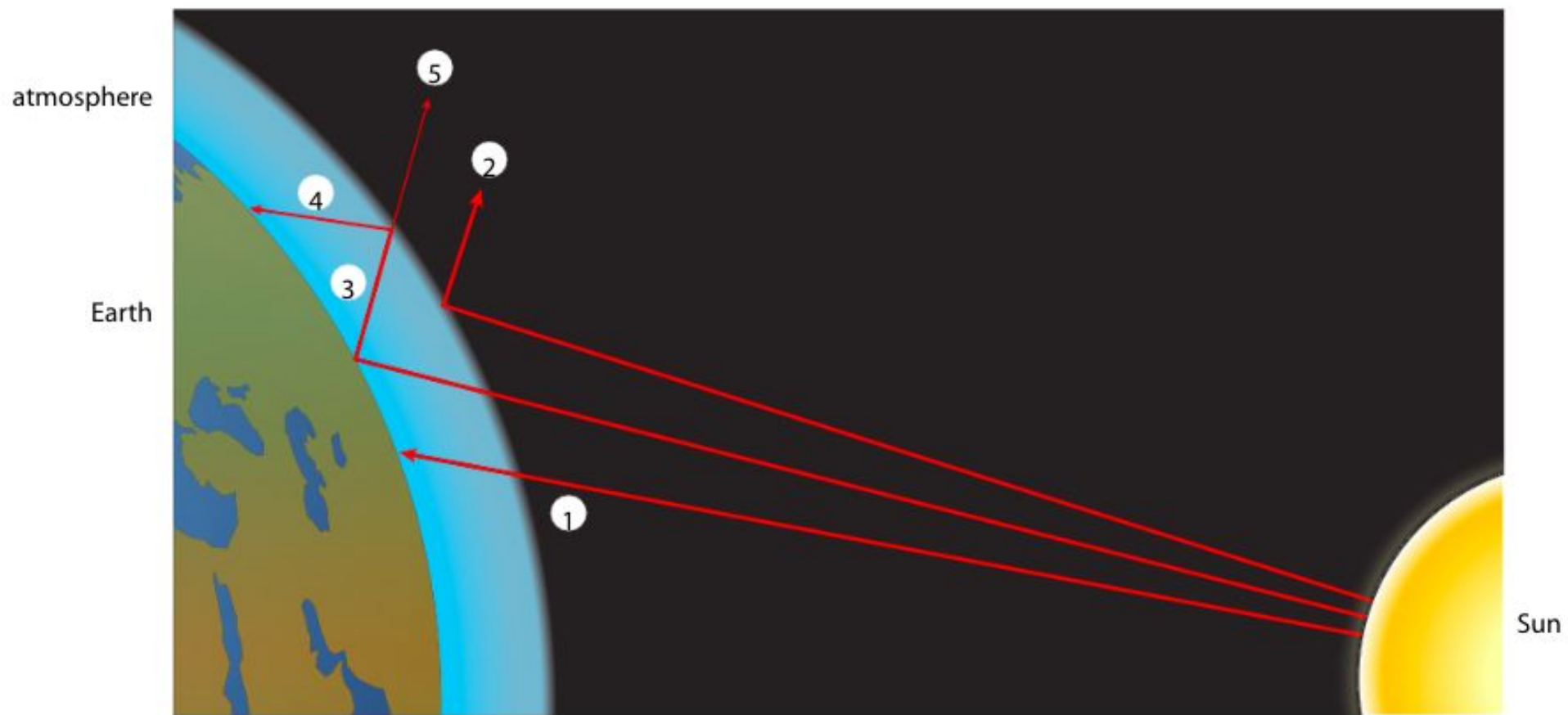
Reader 3:

산업혁명 이후 우리는 대기 중으로 더 많은 온실가스를 배출해 왔습니다. 이러한 배출의 주요 원인은 화석 연료 연소로 인한 이산화탄소와 가축에서 발생하는 메탄입니다.



Reader 4:

삼림 벌채는 문제를 더욱 악화시키고 있습니다. 삼림 벌채는 사람들이 농지, 도로, 건물을 짓기 위해 나무를 베어내는 것을 말합니다. 이는 광합성을 통해 대기 중 이산화탄소를 흡수할 나무가 줄어든다는 것을 의미합니다. 나무가 타거나 썩으면, 나무에 저장된 이산화탄소가 대기 중으로 방출됩니다.



1. Energy from the sun falls on the Earth and warms it up.
 2. Some of this energy does not get through the atmosphere and is reflected back into space.
 3. Energy leaves the Earth and passes into the atmosphere.
 4. Much of this energy is reflected back to Earth by the layer of greenhouse gases such as carbon dioxide, methane and water vapour.
 5. Some of this energy from the Earth passes through the atmosphere into space.
-

Task

1. There are total of 4 worksheets -
Keywords , Task boxes,
Comprehension and Numeracy task
2. You may complete them in any order .
3. I will discuss the questions at the end
of the timer



Keywords

10:00

Match each keyword to its correct meaning, then write down the keyword with its definition.

Emission

An increase in the average temperature on Earth.

Deforestation

Absorbing radiation and radiating it back toward the Earth.

Radiation

Cutting down and/or burning forests.

Greenhouse Effect

High energy waves or particles like infrared light from the Sun.

Global Warming

Producing something and letting it out (especially gas).

Extension: Suggest reasons why changing weather patterns might be a problem.

Keywords (Answers)

Match each keyword to its correct meaning, then write down the keyword with its definition.

Emission

Producing something and letting it out (especially gas).

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Cutting down and/or burning forests.

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An increase in the average temperature on Earth.

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Task Boxes

10:00

Choose at least **two** of the task boxes below to complete.

Questions

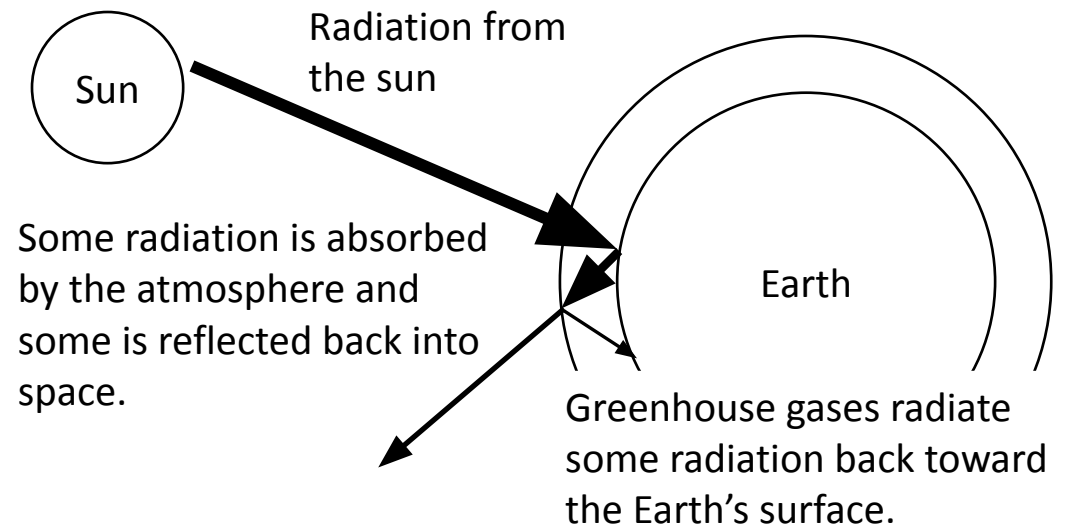
1. Why did the concentration of CO₂ stay the same before 1800?
1. Why did global temperatures start to rise after the Industrial Revolution?

Copy and Fill in the Blanks

Every year more carbon dioxide is _____ than is removed because of humans burning _____ for energy. This causes the _____ of carbon dioxide to increase, leading to _____.

produced **concentration**
fossil fuels **global warming**

Sketch and Label



Challenge Question

Explain how deforestation is contributing to climate change.

Task Boxes (Answers)

Choose at least **two** of the task boxes below to complete.

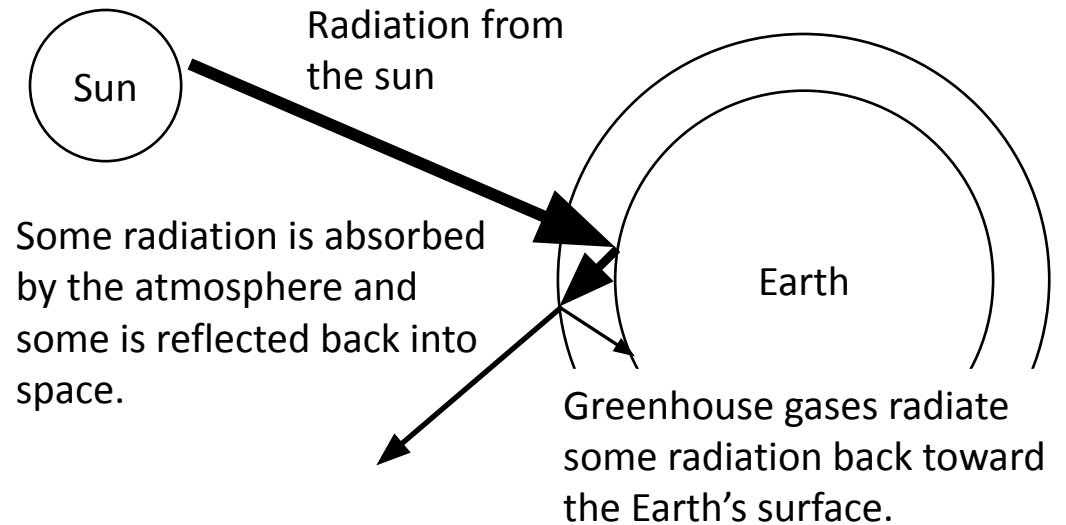
Questions

1. Before 1800 we were not burning fossil fuels for energy.
1. After the Industrial Revolution we started burning fossil fuels releasing greenhouse gases. Modern farming methods also meant more methane from cows and sheep.

Copy and Fill in the Blanks

Every year more carbon dioxide is **produced** than is removed because of humans burning **fossil fuels** for energy. This causes the **concentration** of carbon dioxide to increase, leading to **global warming**.

Sketch and Label

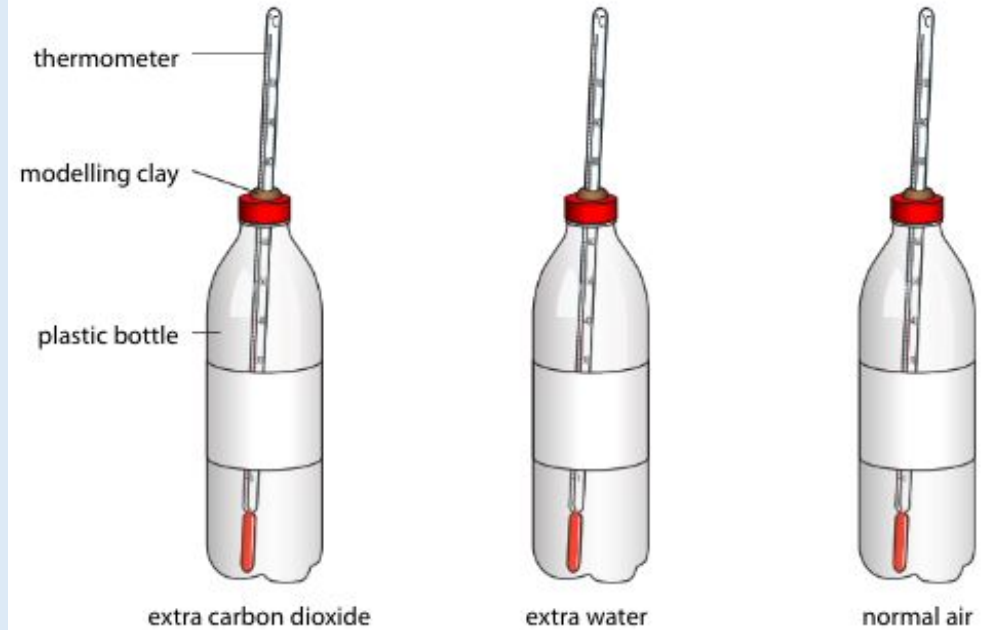


Challenge Question

Deforestation means there are less trees removing CO_2 from the atmosphere. If the trees are burned or rot, this also releases CO_2 .

You will need:

- three large plastic bottles, such as 2-litre drinks bottles with lids, with a hole in each lid
- three thermometers
- a means of fixing the thermometers in place, such as modelling clay
- carbon dioxide supply



Learner's book page 183

Method

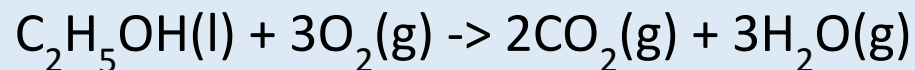
- 1 Place a thermometer in each of the bottles so that the bulb does not touch the bottle in any way.
- 2 Add some carbon dioxide to one bottle.
- 3 Add about 5 cm³ of water to another bottle.
- 4 Leave the third bottle with normal air.
- 5 Label the bottles.
- 6 Place the bottles alongside one another outside. If this is not possible, you can leave them in the classroom near the window.
- 7 Take the temperatures in each bottle at the start.
- 8 Record the temperatures in the three bottles over the next few days. It is up to you to decide when and how often.

Comprehension Questions

10:00

Answer the questions in your book. If you get stuck, try the next one!

1. What do the symbols (l) and (g) mean in this equation?



1. Sketch 3O_2

1. This car is being filled with petrol (a fossil fuel).

Where do fossil fuels come from?



4. Copy and delete the incorrect words:

Greenhouse gases in the atmosphere

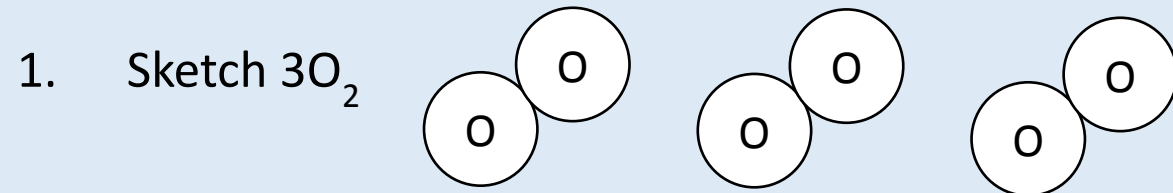
absorb/create some radiation so that it does not go back into **space/the sea**. This is called the **tectonic/greenhouse** effect. It keeps the Earth **colder/warmer** than it would be if all of the radiation went back into space.

Extension: Climate change may result in some species becoming extinct. Explain how one species in an ecosystem dying out can impact on all the other species.

Comprehension Questions (Answers)

Answer the questions in your book. If you get stuck, try the next one!

1. (l) = liquid and (g) = gas



1. Fossil fuels form from the remains of plants and animals over millions of years.



4. Copy and delete the incorrect words:

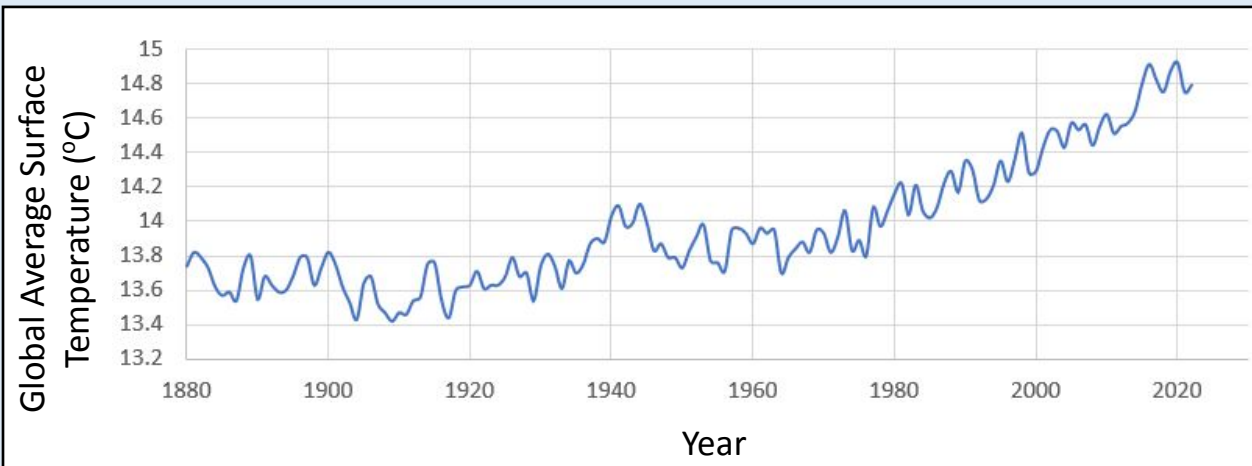
Greenhouse gases in the atmosphere **absorb/create** some radiation so that it does not go back into **space/the sea**. This is called the **tectonic/greenhouse** effect. It keeps the Earth **colder/warmer** than it would be if all of the radiation went back into space.

Extension: Climate change may result in some species becoming extinct. Explain how one species in an ecosystem dying out can impact on all the other species.

Numeracy Task

10:00

Answer the questions in your book. If you get stuck, try the next one!



1. Describe how the global average surface temperature has changed between 1970 and 2020.

2. Melting polar ice is making sea levels rise. What might happen as a result of sea levels rising?

3. Describe how the global average surface temperature changed between 1880 and 1920.

Year	Temperature (°C)
1900	13.82
1925	13.68
1950	13.73
1975	13.89
2000	14.29

4. This table shows the global average temperature every 25 years. In which 25 year period did temperature increase the most?

5. Suggest how scientists could collect data on global temperature. What would they need to control for?

Extension: Scientists predict that by 2100 the temperature could increase by up to 5°C. Suggest why it is difficult to predict *exactly* how much global temperatures will rise.

Numeracy Task (Answers)

Answer the questions in your book. If you get stuck, try the next one!



1. Between 1970 and 2020 global average temperature has increased.

2. Sea levels rising will mean flooding of coastal areas and low-lying land.

3. Between 1880 and 1920, there was little change in average temperature.

Year	Temperature (°C)
1900	13.82
1925	13.68
1950	13.73
1975	13.89
2000	14.29

4. The average global temperature increased the most between 1975 and 2000.

5. Scientists could measure the temperature with a thermometer. They would need to measure the same places at the same times of year.

Extension: Scientists predict that by 2100 the temperature could increase by up to 5°C. Suggest why it is difficult to predict *exactly* how much global temperatures will rise.

Starter - MCQ check



1. Which human activity releases the MOST greenhouse gases?
 - A) Recycling
 - B) Transport
 - C) Burning fossil fuels for electricity
2. What happens when too many greenhouse gases trap heat?
 - A) Ozone layer grows
 - B) Global temperatures rise
 - C) Earth becomes colder
3. Which of these is NOT a consequence of global warming?
 - A) Rising sea levels
 - B) More droughts
 - C) Lower carbon dioxide levels

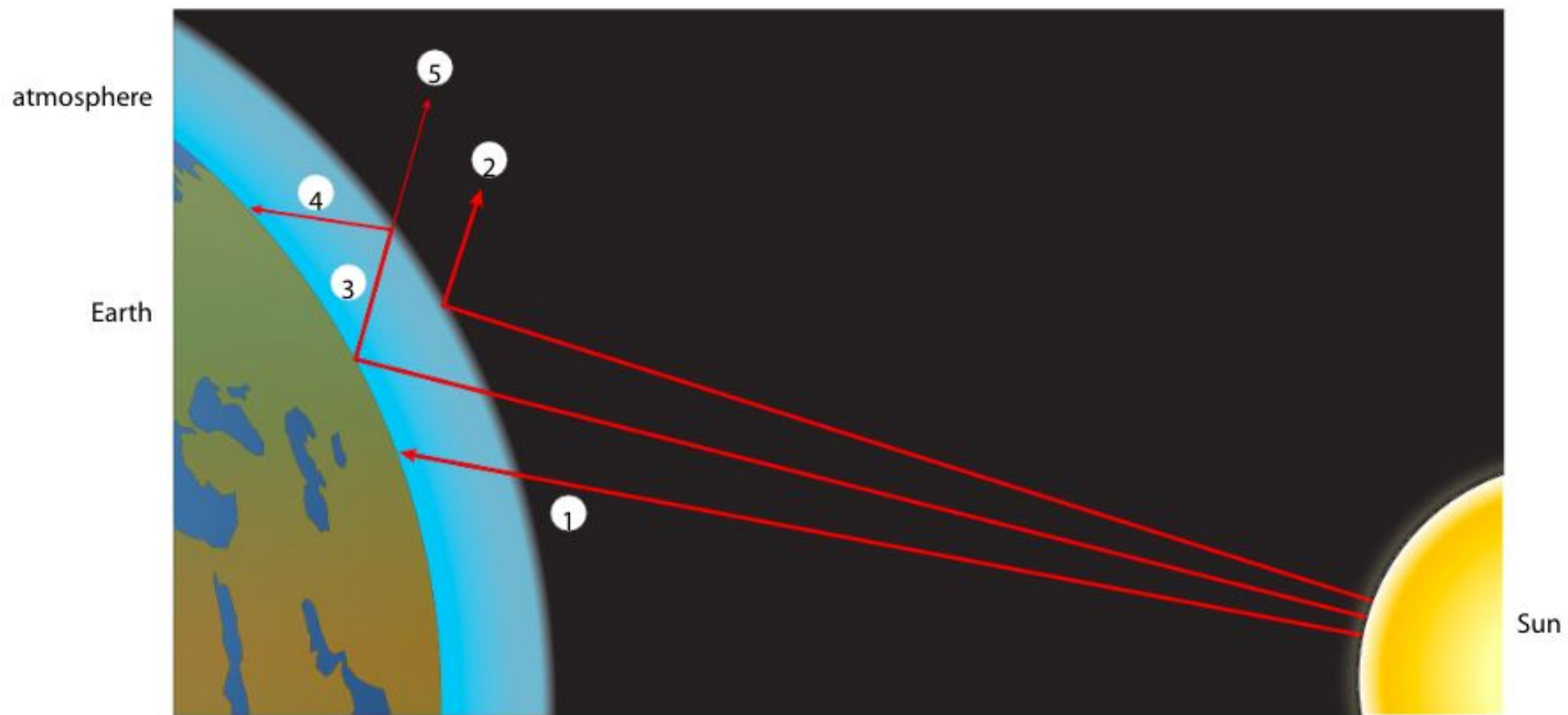
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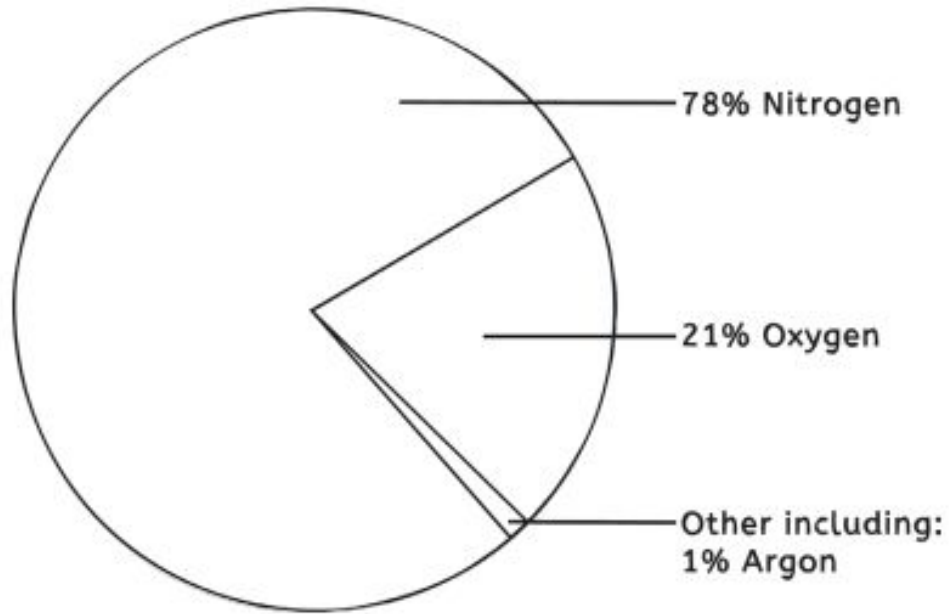
Success criteria

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-

10:00



1. The pie chart above shows the composition of the Earth's atmosphere. Show the data in a suitable table below. [3]

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Table to show the percentage of gases in the atmosphere

Gas	% of atmosphere
Nitrogen	78
Oxygen	21
Other	1

1 mark each for: title, headings, correct data

2. Draw a circle around the activities that increase the level of carbon dioxide in the atmosphere. [2]

burning fossil fuels

driving vehicles

deforestation

recycling materials

3. Name two greenhouse gases. [2]

Any two from: water vapour, carbon monoxide, carbon dioxide, methane, nitrous oxide, CFCs (chlorofluorocarbons), ozone

4a. Describe what the graph is showing. [2]

gradual increase over time / increase in sea level / rising sea levels over time

4b. Suggest a possible explanation for the changes shown on the graph, and describe the impact these changes will have on Earth and humans. [6]

Max six marks from:

- **climate change**
- **global temperature increase / global warming**
- **melting of polar ice caps / glaciers and ice melt**
- **sea levels rise**
- **coastal areas flood**
- **global warming due to greenhouse gas increase**
- **increase of CO₂ in atmosphere**
- **sun's energy trapped by atmosphere**
- **radiation trapped by gases**



TASK:

40:00

Create a leaflet for the general public explaining what global warming is and how humans are contributing towards it.

EXTENSION: Produce a Frequently asked questions section, giving answers for commonly asked questions about climate change.



Global Warming Feedback

WWW

- Is the information **clear** and **concise**?
- Is it well **presented**?
- Have the picked out the **key** information?
- If you didn't know anything about global warming and read this leaflet, how would you **feel**?

EBI

- Any key information **missing**?
- How is the **layout**?
- How is the **presentation**?



10:00

Extra time?

Workbook page 97-103

